Noah Darwich | ML Engineer & Research Data Scientist

Essen, Germany · darwich.noah.95@gmail.com · +49 178 515 55 35 · in/noah-darwich · www.noahdarwich.com/

SUMMARY

ML Engineer and Research Data Scientist with experience building AI/ML systems, multilingual datasets, and research automation tools. Most recently built STT, LLM and TTS pipelines for AI-powered voice agents, with a background spanning machine learning, data engineering and computational social science. Experienced in Python, LLM APIs, NLP, machine learning and large-scale data workflows across academic and applied research environments.

EXPERIENCE

ML Engineer
AethexAI **April 2026 - October 2026, Remote**

- Built and maintained multilingual STT, LLM and TTS pipelines for AI-powered voice agents.
- Developed, fine-tuned and evaluated speech and language models for African and Arabic markets, covering dialect-rich English, French and Arabic.
- Built automated evaluation pipelines and MLOps workflows to improve model quality, reliability and deployment.
- Integrated AI models into production voice-agent systems and supported scalable inference workflows.

Research Data Scientist
Freelance **April 2025 - April 2026**

- **Yale University Collaboration:**
- Designed and developed a custom AI-powered data collection tool using Python, SQL, and LLM APIs to automate protest event extraction from unstructured text sources, processing around 100,000 articles.
- Collaborated with a research team to compile and structure protest event data across 15 sources for research on political mobilization in Jordan.
- **Oxford University Collaboration:**
- Conducted a systematic data landscape assessment for a pilot conference on exploitation and forced labor in market systems, identifying gaps across more than 20 existing datasets.
- Synthesized qualitative and quantitative research findings to evaluate the current state of modern slavery data collection methodologies.

Research Data Scientist
Rockwool Funden **December 2024 - August 2025, Berlin, Germany**

- Designed and executed quantitative public policy research analyzing the relationship between infrastructure changes (e.g. hospital closures) and German electoral voting patterns.
- Cleaned, preprocessed, and merged multiple public datasets using Python, ensuring data integrity for regression analysis.
- Collaborated with an interdisciplinary team of economists and policy researchers to develop research methodology and validate statistical approaches.

Project Manager
Georgetown University **April 2023 - October 2024, Washington, D.C., USA**

- Led a multi-country research project developing quantitative protest event datasets for the 2019 Arab Spring movements in Sudan, Algeria, and Lebanon.
- Managed 3 research teams totaling 10 assistants, overseeing project timelines, deliverables, and quality assurance.
- Designed and delivered training programs on data collection methodology and coding protocols for research staff.
- Developed automated research tools (i.e., Python scripts, AI models) to streamline protest event data collection by identifying events, categorizing, and storing key variables from Arabic news articles for large-scale research.

Senior Research Assistant
Georgetown University **January 2021 - July 2023, Washington, D.C., USA**

- Developed algorithmic data collection methods in Python to automate key variable extraction and data entry.
- Collected and analyzed protest event data from the Iraqi Arab Spring protest wave (2011) and the 2019 Tishreen uprising, drawing on local Iraqi news sources and social media sources.
- Collected Arabic-language qualitative data on Egypt's football ultras; analyzed data on home districts of ultras killed during and after the 2011 revolution.

Data Science Lecturer
Le Wagon **April 2023 - July 2024, Cologne, Germany**

- Taught data science fundamentals, including Python, statistical analysis, machine learning algorithms, and data visualization, to bootcamp students.
- Led hands-on workshops covering Python, SQL, pandas, NumPy, scikit-learn, and deep learning frameworks.
- Supervised capstone projects, providing technical guidance on model selection, feature engineering, and performance optimization.

Senior Teaching Assistant
Le Wagon **July 2022 - April 2023, Cologne, Germany**

- Provided technical support to students in the full-time Data Science bootcamp, helping them complete their daily coding challenges, from debugging their code to reinforcing fundamental concepts from their coursework.

Research Assistant
Princeton University **April 2018 - October 2021, New Jersey, USA**

- Constructed a quantitative protest event dataset covering Egypt's political transition (2012–2013), extracting and coding events from Arabic-language newspaper al-Masry al-Youm.
- Conducted descriptive statistical analysis to identify mobilization trends and patterns across event types, locations, and participant demographics.
- Performed data validation and diagnostic assessments to ensure coding accuracy and dataset reliability.

Research Assistant
American University of Beirut **January 2019 - May 2021, Beirut, Lebanon**

- Compiled a longitudinal protest event dataset spanning 30 years (1989–2019) of Lebanese social movement activity, coding events from Arabic and English-language sources.
- Analyzed historical mobilization patterns and trends, coding events across variables including protest type, geographic location, and participant demands.

Research Assistant
Princeton University **November 2019 - May 2020, New Jersey, USA**

- Contributed to the construction of a mixed-methods dataset combining ethnographic fieldwork with quantitative analysis for research on informal financial networks in wartime Syria "How money moves in wartime Syria" (winner of the 2021 ASA Best Dissertation Award and the 2021 MESA Malcolm Kerr Best Dissertation Award).

PROJECT

Automating Data Collection
January 2026

- Developed an end-to-end automated pipeline for protest event data collection using Python, LLMs, and the OpenAI API.
- The tool extracts over 80 key variables (e.g., location, date, event type, demands, participants) from Arabic-language news articles.
- Deployed scalable cloud infrastructure on GCP to handle batch processing of large document collections and API requests.
- Designed cloud-based data storage and retrieval system (PostgreSQL) with automated validation checks ensuring data integrity.

PRO-TEST
Le Wagon Graduation Project **July 2022**

- Developed an end-to-end ML pipeline predicting protest violence, from raw data ingestion to deployed web application with 80% classification accuracy.
- Engineered 100+ categorical features from 13,000 event records using one-hot encoding, target encoding, and correlation analysis for feature selection.
- Built and optimized a Random Forest model using GridSearchCV/RandomizedSearchCV, evaluating performance with precision, recall, and F1 metrics.
- Designed and deployed a Streamlit web app, enabling users to input event characteristics and receive real-time risk predictions.

EDUCATION

Bachelor of Business Administration
Tishreen University · Latakia, Syria · 2018 · 85%

Data Science Bootcamp
Le Wagon · Cologne, Germany · 2022

- Advanced Python & SQL for Data Science.
- Data Analysis & Visualization.
- Un/supervised Machine Learning.
- Deep Learning & Neural Networks.
- Data Engineering.

COURSEWORK

Data Analysis with Python
Freecodecamp.com · 2022 · Python, NumPy, Pandas, Matplotlib, Seaborn

- Reading data from multiple sources
- Process and visualize data
- Using NumPy, Pandas, Matplotlib and Seaborn libraries.

INVOLVEMENT

Volunteer Interpreter
Istanbul, Turkey · Safe place Istanbul · June 2018 - August 2019

- An NGO that provides services for LGBTQ+ refugees.

SKILLS

Machine Learning - NLP - LLMs - RAG - Model Fine-tuning - AI Automation
Speech & Voice AI - ASR/STT - TTS - Real-time Voice Pipelines - Telephony Integration
MLOps - Model Evaluation - LLM Observability - Model Serving & Inference
Python - SQL - Git - REST APIs - Cloud Computing - Databases
PyTorch - TensorFlow - Scikit-learn - Pandas - NumPy - Data Visualisation
Data Analysis - Quantitative Research - Team Management - Data Quality Control - Computational Social Science
Arabic - Native · English - Fluent · German - Fluent · Turkish - Conversational